Sibley School of Mechanical and Aerospace Engineering

MAE 3240 Heat Transfer

Spring 2026

Problem Set #5

Posted: Monday, Mar 2 **Due: Tuesday, Mar 10, 11:59 pm**

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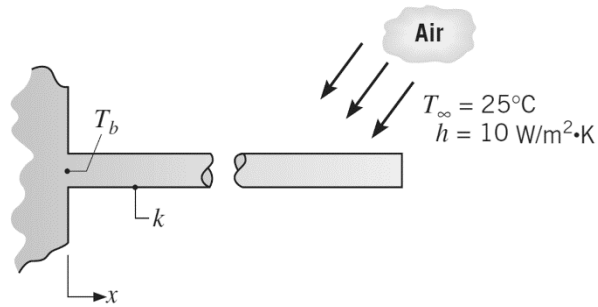
1. (60 pt) A cylindrical pin fin with a radius of r = 1 mm and length of L = 5 cm is exposed to airflow with a temperature of T_∞ = 25 °C and a heat transfer coefficient of h = 10 W/m^2·K. Thermal conductivity of the pin fin is k = 200 W/m·K. The tip of the pin fin is thermally insulated. Use the fin problem solving workflow discussed in the lecture to solve the following questions.



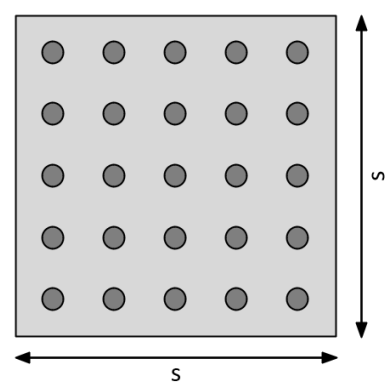
   (a) If the fin base temperature is T_b = 125 °C, please calculate the fin temperature at $x = L/2$ and $x = L$.

   (b) Calculate the fin effectiveness ε_f and the fin efficiency η_f based on their definition and results provided by Table 3.4 of the textbook.

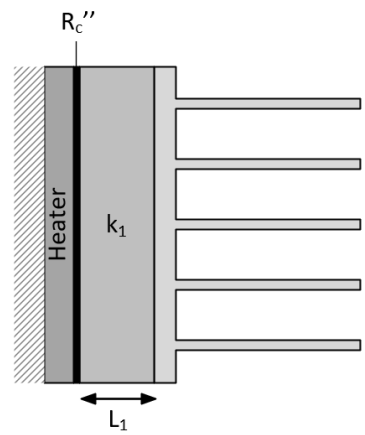
   (c) Calculate the fin efficiency η_f based on results provided by Table 3.5. Compare the results of (b) and (c) and discuss if Table 3.5 provided a good approximation to the fin efficiency.

   (d) Calculate the thermal resistance of this pin fin R_f.

   (e) The pin fins then form a 5×5 array mounted on a base plate with a side length of s = 3 cm, as shown in the figure on the right-hand side. Calculate the overall efficiency of the fin array η_o and the thermal resistance of the fin array R_o. Calculate how much heat is dissipated through the unfined surface and how much is through the fins.



(f) The heat sink consisting of this fin array is used to improve the heat dissipation from a thin film heater, as shown in the figure on the right-hand side. The heat sink is mounted on a slab with thickness L_1 = 2 cm and thermal conductivity k_1 = 10 W/m·K. There is a thermal adhesive between the thin film heater and the slab, resulting in a contact thermal resistance R_c = 10^{-4} m^2·K/W. Thermal insulation was applied to the backside of the heater. The heater generates a uniform heat flux q'' = 1000 W/m^2, which was dissipated through the heat sink. The temperature across the thin film heater is uniform. Thermal resistance for the base of the heat sink can be neglected. There is no contact thermal resistance between the base of the heat sink and the slab. Please draw the thermal resistance network of this system and calculate the temperature of the thin film heater T_h. Compared with the condition without the heat sink, *i.e.*, directly exposing the heater and the slab to ambient air, how much temperature reduction of the heater is achieved?



2. (30 pt) A long wire of diameter D = 1 mm is submerged into an oil bath of temperature T_∞ = 20 °C. Density ρ, thermal conductivity k, and heat capacity c of the wire are 4000 kg/m^3, 400 W/m·K, and 500 J/kg·K, respectively. The wire has an electrical resistance per unit length of R_e' = 0.01 Ω/m. The oil bath can dissipate heat with a heat transfer coefficient of h = 500 W/m^2·K. At the initial moment t = 0 s, a current of I = 100 A flows through the wire and produces Joule heating.

   (a) Calculate the steady-state temperature of the wire.

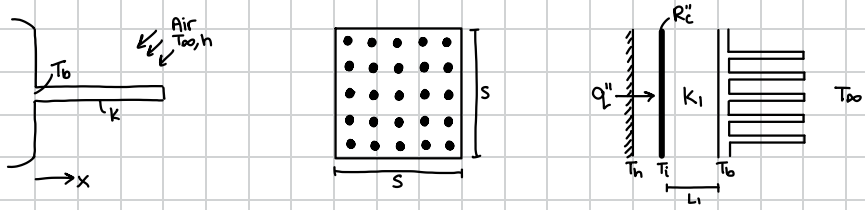
   (b) Apply the lumped capacitance method to derive an expression for the wire temperature as a function of time. How long does it take for the wire to reach a temperature that is within 5 °C of the steady-state temperature?

Problem 1

Knowns: cylindrical pin fin, $r = 1mm$, $L = 5cm$, $T_\infty = 25°C$, $h = 10 W/m^2K$, $k = 200 W/mK$, tip condition: thermally insulated
$T_b = 125°C$, $S = 3cm$, $L_1 = 2cm$, $k_1 = 10 W/mK$, $R_c'' = 10^{-4} m^2K/W$, backside of heater insulated, $q'' = 1000 W/m^2$

Find: a) $T(x=L/2)$, $T(x=L)$ b) ε_f, η_f 3.4 c) η_f 3.5 d) R_f e) η_o, R_o, q_{uf}, q_f f) thermal resistance network, T_b, η_o

Schematic:



Assumptions: 1) 1D heat transfer, 2) uniform heat flux through heater, 3) neglect thermal resistance of base, 4) no contact thermal resistance b/w heat sink base and slab, 5) tip area ≪ surface area, 6) neglect thermal radiation

Analysis:

a)
$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{(10)(\pi)(0.002)}{(200)(\pi)(0.001)^2}} \Rightarrow m = 10$

Table 3.4: $\theta(x=L) = \frac{\cosh[m(L-x)]}{mL}\Big|_{x=L}\theta_b = \frac{\cosh(0)}{\cosh[(10)(0.05)]}(125-25) \Rightarrow \theta(x=L) = 88.68$

$T(x=L) = \theta(x=L) + T_\infty = 86.68 + 25 \Rightarrow T(x=L) = 113.9\ °C$

$\theta(x=L/2) = \frac{\cosh[10(0.05-0.025)]}{\cosh[10(0.05)]}(125-25) \Rightarrow \theta(x=L/2) = 91.47\ °C$

$T(x=L/2) = 91.47 + 25 \Rightarrow T(x=L/2) = 116.47\ °C$

b)
Table 3.4: $q_f = M\tanh(mL)$

$M = \sqrt{hPkA_c}\,\theta_b = \sqrt{(10)(\pi)(0.002)(200)(\pi)(0.001)^2}(125-25) \Rightarrow M = 0.628$

$q_f = (0.628)\tanh[(10)(0.05)] \Rightarrow q_f = 0.290\ W$

$q_{nf} = hA_c\theta_b = (10)(\pi)(0.001)^2(125-100) \Rightarrow q_{nf} = 0.00314\ W$

$\varepsilon_f = \frac{q_f}{q_{nf}} = \frac{0.290}{0.00314} \Rightarrow \varepsilon_f = 92.4$

$\eta_f = \frac{q_f}{hA_f\theta_b} = \frac{0.290}{(10)(\pi)(0.002)(0.05)(125-25)} \Rightarrow \eta_f = 0.923$

c)
Table 3.5: $\eta_f = \frac{\tanh(mL_c)}{mL_c}$

$L_c = L + \frac{D}{4} = 0.05 + \frac{0.002}{4} \Rightarrow L_c = 0.0505\ m$

$\eta_f = \frac{\tanh[(10)(0.0505)]}{(10)(0.0505)} \Rightarrow \eta_f = 0.923$

Parts b and c produced the same resuts for η_f, so Table 3.5 does provide a good approximation to the fin efficiency

d)
$R_f = \frac{\theta_b}{q_f} = \frac{125-25}{0.290} \Rightarrow R_f = 345\ K/W$

e)
$\eta_o = 1 - \frac{NA_f}{A_{tot}}(1-\eta_f)$

$NA_f = (25)(\pi)(0.002)(0.05) \Rightarrow NA_f = 0.00785$

$A_{uf} = (0.03)^2 - (25)(\pi)(0.001)^2 \Rightarrow A_{uf} = 0.000821$

$A_{tot} = 0.00785 + 0.000821 \Rightarrow A_{tot} = 0.00868$

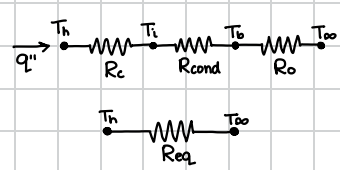
$\eta_o = 1 - \frac{0.00785}{0.00868}(1-0.923) \Rightarrow \eta_o = 0.93$

$R_o = \frac{1}{\eta_o hA_{tot}} = \frac{1}{(0.93)(10)(0.00868)} \Rightarrow R_o = 12.4\ K/W$

$q_{fs} = Nq_f = 25(0.29) \Rightarrow q_{fs} = 7.25\ W$

$q_{uf} = hA_{uf}\theta_b = (10)(0.000821)(125-25) \Rightarrow q_{uf} = 0.821\ W$

f)



$R_{eq} = R_c + R_{cond} + R_o = \frac{R_c''}{A} + \frac{L_1}{k_1A} + R_o$

$= \frac{10^{-4}}{0.03^2} + \frac{0.02}{(10)(0.03)^2} + 12.4 \Rightarrow R_{eq} = 14.72\ K/W$

$\frac{T_h - T_\infty}{R_{eq}} = q \rightarrow T_h = q''AR_{eq} + T_\infty = (1000)(0.03)^2(14.72) + 25$

$T_h = 38.3\ °C$

Without heat sink:

$\frac{T_i - T_b}{R_{cond}''} = q''$

$R_{cond}'' = \frac{L_1}{k_1} = \frac{0.02}{10} \Rightarrow R_{cond}'' = 0.002$

$T_i = q''R_{cond}'' + T_b = (1000)(0.002) + 125 \Rightarrow T_i = 127\ °C$

$\frac{T_h - T_i}{R_c''} = q''$

$T_h = q''R_c + T_i = (1000)(10^{-4}) + 127 \Rightarrow T_h = 127.1$

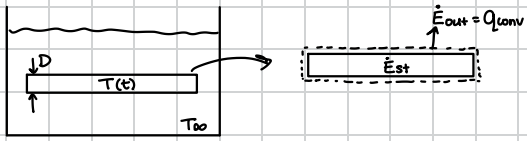
$127.1 - 38.3 = 88.8\ °C$ temp reduction

Problem 2

Knowns: long wire, $D = 1mm$, $T_\infty = 20\,°C$, $\rho = 4000\,kg/m^3$, $k = 400\,W/mK$, $c = 500\,J/kgK$, $R_e' = 0.01\,\Omega/m$, $h = 500\,W/m^2K$, $I(t=0) = 100A$

Find: a) T_{ss} b) $T(t)$, t for a temp within 5°C of T_{ss}

Schematic:



Assumptions: 1) neglect temp gradients within the wire, 2) 1D heat transfer, 3) no heat generation (part b), 4) transient conduction, 5) neglect radiation

Analysis:

a) Energy balance: $\cancel{\dot{E}_{in}}^{\,0} - \dot{E}_{out} + \dot{E}_g = \dot{E}_{st} \longrightarrow -\dot{E}_{out} + \dot{E}_g = \dot{E}_{st} \longrightarrow -hA_s(T_s - T_\infty) + I^2R_e = \rho V c \frac{dT}{dt}$

Steady state: $\frac{dT}{dt} = 0 \longrightarrow hA_s(T_s - T_\infty) = I^2R_e \longrightarrow h\pi DL(T_s - T_\infty) = I^2R_e \longrightarrow h\pi D(T_s - T_\infty) = I^2R_e'$

$T_{ss} = \frac{I^2R_e'}{h\pi D} + T_\infty = \frac{(100)^2(0.01)}{(500)(\pi)(0.001)} + 20\,°C \Longrightarrow T_{ss} = 83.7\,°C$

b) $-hA_s(T_s - T_\infty) + I^2R_e = \rho V c \frac{dT}{dt}$

$-h\pi D(T - T_\infty) + I^2R_e' = \rho A_c c \frac{dT}{dt} = \rho\pi(\frac{D}{2})^2 c \frac{dT}{dt}$

$\theta = T - T_\infty \qquad \theta_{ss} = T_{ss} - T_\infty = \frac{I^2R_e'}{h\pi D}$

$-h\pi D(T - T_\infty) = \rho\pi(\frac{D}{2})^2 c \frac{dT}{dt} - I^2R_e'$

$T - T_\infty = \frac{-\rho Dc}{4h}\frac{dT}{dt} + \frac{I^2R_e'}{h\pi D}$

$\theta = \frac{-\rho Dc}{4h}\frac{dT}{dt} + \theta_{ss}$

$\theta_{ss} - \theta = (\frac{\rho Dc}{4h})\frac{dT}{dt}$

$T_{ss} - T_\infty - T + T_\infty = T_{ss} - T$

$T = (\frac{-\rho Dc}{4h})\frac{dT}{dt} + T_{ss}$

$\frac{dT}{dt} + \underbrace{\frac{4h}{\rho Dc}}_{a} T - \underbrace{\frac{4h}{\rho Dc} T_{ss}}_{b} = 0$

$T(t) = T_i e^{-at} + \frac{b}{a}(1 - e^{-at})$

$T(t) = T_\infty \exp[\frac{-4h}{\rho Dc}t] + \frac{4hT_{ss}}{\rho Dc} \cdot \frac{\rho Dc}{4h}(1 - \exp[\frac{-4h}{\rho Dc}t]) = T_\infty \exp[\frac{-4h}{\rho Dc}t] + T_{ss}(1 - \exp[\frac{-4h}{\rho Dc}t])$

$\frac{-4h}{\rho Dc} = \frac{-4(500)}{(4000)(0.001)(500)} = -1$

$T(t) = 20e^{-t} + 83.7(1 - e^{-t}) \Longrightarrow T(t) = 83.7 - 63.7e^{-t}$

$78.7 = 83.7 - 63.7e^{-t}$

$5 = 63.7e^{-t}$

$0.0785 = e^{-t}$

$\ln(0.0785) = -t \Rightarrow t = 2.54s$